

LESSON 10.2

IDENTIFYING DOMAIN AND RANGE



LESSON INTRODUCTION

MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.

LEARNING TARGETS

- I can relate domain and range to independent and dependent variables in a relation.
- I can identify the domain and range of a relation.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can you use the independent and dependent variables to determine the domain and range of a relation?
- How do you identify the domain and range of a relation?
- How do you decide whether to express the domain and range with lists or with compound inequalities?

LESSON NARRATIVE

Students write the domains and ranges of relations and make connections to the independent and dependent variables. Given different representations of relations (ordered pairs, tables, graphs, mapping diagrams), students write the domain and range as a finite list of numbers or as inequalities when the set is infinite.

COMPONENT SUMMARY

10.2.1 WARM-UP (~5 MIN.)

Determine which graph is the best representation

10.2.2 GUIDED INSTRUCTION (35-40 MIN.)

Write the domain and range of a relation, examining why values are not included

10.2.3 TRY IT (5-10 MIN.)

Identify the domain and range given different representations of a relation

10.2.4 WRAP-UP (~5 MIN.)

Reflect on topics they are unsure about and develop a plan to fill in this gap in their understanding

RESOURCES

GLOSSARY

Key Terms:

- Domain
- Range
- Compound Inequality

Additional Terms:

- Relation

MATERIALS

- Highlighters
- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think that a number divided by zero is zero.
- Students may confuse the domain and the range.
- Students may struggle to recognize which values are not included in the domain and range.
- When writing inequalities, students may select the incorrect inequality symbol.

THINGS TO CONSIDER

- Consider creating reference cards or anchor charts with definitions and examples of domain, range, and inequalities.
- Students may need help determining whether to use lists or inequalities to represent the domain and range.

LITERACY AND LANGUAGE ROUTINES

Stronger and Clearer Each Time

In question 5b on 10.2.2 Guided Instruction, students are asked to explain why they represent the domain and range with lists and with compound inequalities. Have students first write their responses individually. Next, pair students up and have them swap responses. Ask students to provide each other with feedback and then give them an opportunity to revise their responses. Throughout this process, students should be pressed for details and encouraged to press each other for details.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

In the 10.2.1 Warm-Up, students connect graphs to the everyday context of the price of frozen yogurt based on the weight of the yogurt. The relationship between the independent variable, the weight of the yogurt, and the price is represented on four graphs. Students determine which graph is the best representation of the relationship.

DIFFERENTIATE INSTRUCTION

Allow students to create and use reference cards with definitions and examples of how to identify the domain and range in the variety of different representations of relations (ordered pairs, tables, graphs, mapping diagrams, and a given rule).

The 10.2.1 Warm-Up provides a visual entry point for students as they determine which graph is the best representation of a scenario. A kinesthetic entry point can be fostered by having students use their finger to trace over how the price changes as the ounces of yogurt increases.

In each representation of the relation, have students highlight the elements of the domain and range with two different colors. For example, the elements of the domain could be highlighted in pink and the elements of the range could be highlighted in yellow. This intermediate step, before students express the domain and range with lists or compound inequalities, will help students organize the values and avoid making careless mistakes.

10.2.1 WARM-UP: MATH TALK

DIFFERENTIATE INSTRUCTION

If students struggle getting started, have them first determine the price for 2 ounces, 7 ounces, and 10 ounces based on the pricing policy. Then have the students find the graph that matches these three prices for these serving sizes.

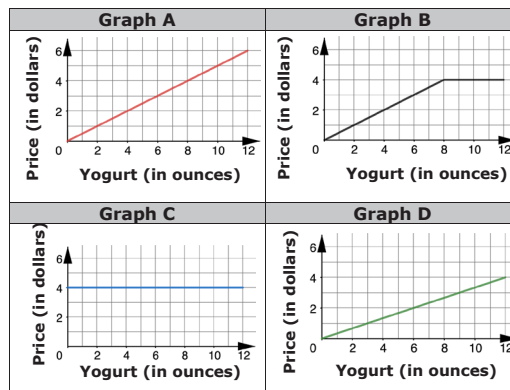
A kinesthetic entry point can be fostered by having students use their finger to trace over how the price changes (increases, decreases, remains constant) as the weight of the yogurt increases.



10.2.1 Warm-Up: Math Talk

A self-serve frozen yogurt store sells servings up to 12 ounces. It charges \$0.50 per ounce for a serving between 0 and 8 ounces, and \$4.00 for any serving greater than 8 ounces and up to 12 ounces.

Four graphs are shown.



- Explain which graph best represents the price as a function of the weight of a serving of yogurt.
Graph B best represents the price as a function of the weight of a serving of yogurt. Graph B shows the yogurt price increasing by .5 increments until the serving size is 8 ounces and then remaining at \$4 from 8-12 ounces.



10.2.2 Guided Instruction: Identifying the Domain and Range of a Relation

- Consider the statements shown.

$$12 \div 3 = 4 \text{ because } 12 = 4 \cdot 3.$$

$$6 \div 0 = x \text{ because } 6 = x \cdot 0.$$

In the second statement, what value can be used in place of x so that the statement is true?

Answers will vary. There is no number that can be multiplied by 0 to get something other than 0. It is also not possible to divide a number in 0 groups. Therefore, there is no value of x that makes the statement true. The value of x is undefined.

The inputs of a relation are described as its **domain**.



The **domain** is the complete set of possible values of the input of a function or relation. The domain may vary depending on the context.

Sometimes one or more values are not allowed in the domain.

- Use the rule to complete the table of values to represent the relation described by the rule and answer the subsequent questions.

- Complete the table of values.

Rule A	Table of Values												
$x \Rightarrow$ divide 1 by the input $\Rightarrow \frac{1}{x}$	<table> <tr> <th>Input</th><th>Output</th></tr> <tr> <td>$-\frac{3}{7}$</td><td>$-\frac{7}{3}$</td></tr> <tr> <td>1</td><td>1</td></tr> <tr> <td>$\frac{4}{5}$</td><td>$\frac{5}{4}$</td></tr> <tr> <td>0</td><td>Not possible</td></tr> <tr> <td>12</td><td>$\frac{1}{12}$</td></tr> </table>	Input	Output	$-\frac{3}{7}$	$-\frac{7}{3}$	1	1	$\frac{4}{5}$	$\frac{5}{4}$	0	Not possible	12	$\frac{1}{12}$
Input	Output												
$-\frac{3}{7}$	$-\frac{7}{3}$												
1	1												
$\frac{4}{5}$	$\frac{5}{4}$												
0	Not possible												
12	$\frac{1}{12}$												

10.2.2 GUIDED INSTRUCTION: IDENTIFYING THE DOMAIN AND RANGE OF A RELATION

TEACHER MOVES

Spend time exploring division by zero with students. Represent the division by restating it as dividing into zero groups. The purpose of this question is to lead into the concept of domain restrictions and understanding that in some relations, certain values are not allowed. Revisit this concept in question 2, part a.

DIFFERENTIATE INSTRUCTION

Have students highlight the vocabulary word and the word “input” in the definition of domain. As students engage with these words throughout the lesson, have the students repeat both words. For example, in the question 2b, when students read “the domain does not include zero,” students also say “the input does not include zero.”

TEACHER MOVES

Spend time probing students of their understanding of the phrase “possible values” in the definition of domain and the connection to question 1, which has no possible value for x , and the statement “Sometimes one or more values are not in the domain.” Use questions such as “Are there places you are not allowed to go?” or “Are there people who you try to keep out of your room?” to help students relate to excluding a value or values in a domain.

ENRICHMENT

- How do you determine which values are not allowable?
- Why can’t we divide by zero?

10.2.2 GUIDED INSTRUCTION: IDENTIFYING THE DOMAIN AND RANGE OF A RELATION (CONTINUED)

TEACHER MOVES

Engage students in a discussion about determining which inequality symbol to use and when to use a compound inequality. When is the inequality symbol open to the left versus right? When is the inequality symbol underlined versus not underlined?

DIFFERENTIATE INSTRUCTION

Have students highlight the vocabulary word and the word “output” in the definition of range. As students engage with these words throughout the lesson, have the students repeat both words.

ENRICHMENT

Why do we say “unallowable” when referring to input values but say “unattainable” when referring to output values?

- b. In this relation, 0 is not an allowable input.
Therefore, the domain does not include 0.

- c. Work with your partner to determine if there is another input that is not allowable for $\frac{1}{x}$.
There are no other inputs that are not allowable.

The domain of Rule A can be represented using inequalities for the input values defined as x .

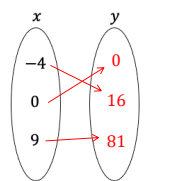
$$x > 0 \text{ or } x < 0$$

These two inequalities can also be written as one, $x \neq 0$.

The outputs of a relation are described as its **range**.

 The **range** is the complete set of possible values of the output of a relation or function.

3. Use the rule to complete the mapping diagram and answer the subsequent questions.
a. Complete the mapping diagram.

Rule B	Mapping Diagram
$x \Rightarrow$ square the input $\Rightarrow y$	

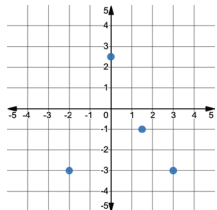
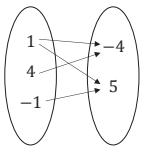
- b. Based on the rule, what type of output values are unattainable for this relation?
Negative Numbers are unattainable for this relation.

- c. Write an inequality to represent the range of this relation.

$$y \geq -0$$

At times, the domain and range of a relation can be represented using a list of numbers.

4. For each relation, write the domain and range as a set of numbers.

Relation A	Relation B
 Domain: $\{-2, 0, 1, 5, 3\}$ Range: $\{-3, -1, 2, 5\}$	 Domain: $\{-1, 1, 4\}$ Range: $\{-4, 5\}$

There are also instances where the domain and range of a relation can be represented by a compound inequality.



A **compound inequality** is a conjunction of two or more inequalities.

For example, $-4 \leq x \leq \frac{3}{5}$ is a compound inequality that is read, “ -4 is less than or equal to x , which is less than or equal to $\frac{3}{5}$.” This inequality is true for any value of x between -4 and $\frac{3}{5}$, including either of those values.

10.2.2 GUIDED INSTRUCTION: IDENTIFYING THE DOMAIN AND RANGE OF A RELATION (CONTINUED)

DIFFERENTIATE INSTRUCTION

Students may struggle with writing the location of these points when they lie between grid lines. Have the students first write the ordered pairs from the graph before writing the domain and range.

ENRICHMENT

How can you use context clues to determine what the word “conjunction” means in the definition of compound inequality?

10.2.2 GUIDED INSTRUCTION: IDENTIFYING THE DOMAIN AND RANGE OF A RELATION (CONTINUED)

DIFFERENTIATE INSTRUCTION

Using a ruler, have the students place the ruler on the leftmost point and highlight on the x-axis the values of that point. Then place the ruler on the rightmost point and highlight on the x-axis this value. Shade in all the values on the x-axis between these points where the graph has a corresponding value. Repeat this with a different color highlighter on the y-axis using the lowest and highest points.

TEACHER MOVES

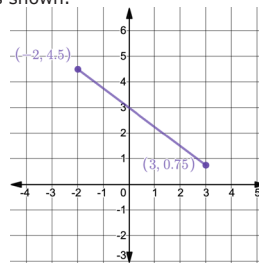
Stronger and Clearer Each Time

Students are asked to explain why they represent the domain and range with lists and with inequalities.

- Have students first write their responses individually.
- Next, pair students up and have them swap responses. Ask students to provide each other with feedback and then give them an opportunity to revise their responses.

Throughout this process, students should be probed for details and encouraged to probe each other for increasingly precise responses.

5. A graph is shown.



- a. Represent the domain and range of the graph using compound inequalities.

Domain: $-2 \leq x \leq 3$

Range: $0.75 \leq y \leq 4.5$

- b. Compare this graph to Relation A in the previous problem. Discuss with your partner why representing the domain and range of this graph could not be done by writing the elements of each in a list.

Answers will vary. In Relation A, there were 4 elements in the domain and range because the relation consisted of 4 points on a graph. The graph shown in question 5 is a line segment meaning it consists of an infinite number of points.



Representing Domain and Range

When the elements of the domain or range consist of a **set number of values**, each can be **represented by a list**.

When the elements of the domain or range consist of an **infinite number of points between specified values**, an inequality is best.



10.2.3 Try It: Identifying the Domain and Range of a Relation

- Identify the domain and range of each relation.
Represent the domain as a list or using an inequality.

Relation	Domain										
<table border="1"> <tr> <th>x</th><th>y</th></tr> <tr> <td>-4.7</td><td>7</td></tr> <tr> <td>2.4</td><td>4.7</td></tr> <tr> <td>-1.6</td><td>3.9</td></tr> <tr> <td>8</td><td>10</td></tr> </table>	x	y	-4.7	7	2.4	4.7	-1.6	3.9	8	10	$\{-4.7, -1.6, 2.4, 8\}$ Range $\{3.9, 4.7, 7, 10\}$
x	y										
-4.7	7										
2.4	4.7										
-1.6	3.9										
8	10										
Relation	Domain										
$\{(2.3, 7.4), (3.8, 6), (1.9, 5.2), (2.3, 8)\}$	$\{1.9, 2.3, 3.8\}$ Range $\{5.2, 6, 7.4, 8\}$										
Relation	Domain										
$x \Rightarrow$ divide 5 by the input $\Rightarrow \frac{5}{x}$	$x < 0$ and $x > 0$ Range $y < 0$ and $y > 0$										
Relation	Domain										
	$-1 \leq x \leq 4$ Range $-9 \leq y \leq 9$										

10.2.3 TRY IT: IDENTIFYING THE DOMAIN AND RANGE OF A RELATION

DIFFERENTIATE INSTRUCTION

In each representation of the relation, have students highlight the elements of the domain and range with two different colors. For example, the elements of the domain could be highlighted in pink and the elements of the range could be highlighted in yellow.

This scaffolding, before students express the domain and range with lists or inequalities, will help students organize the values and avoid making careless mistakes.

TEACHER MOVES

Anticipate students struggling with writing the domain and range for the rule “divide 5 by the input.” Have questions or suggestions prepared to move them forward. For example, “Try making a table of input/output pairs for the rule. Include a variety of rational numbers as inputs. Do you notice any numbers that might not have an output?”



10.2.4 Wrap-Up: Growth Mindset

Complete each statement.

- Today I am still unsure about...
Answers will vary.
- To fill this gap, I intend to...
Answers will vary.

10.2.4 WRAP-UP: GROWTH MINDSET

TEACHER MOVES

Consider prompting students to look back over the topics they studied in the lesson and code the sections based on their understanding of each part. For example, 2 = fully understand, 1 = partially understand, and 0 = do not understand.